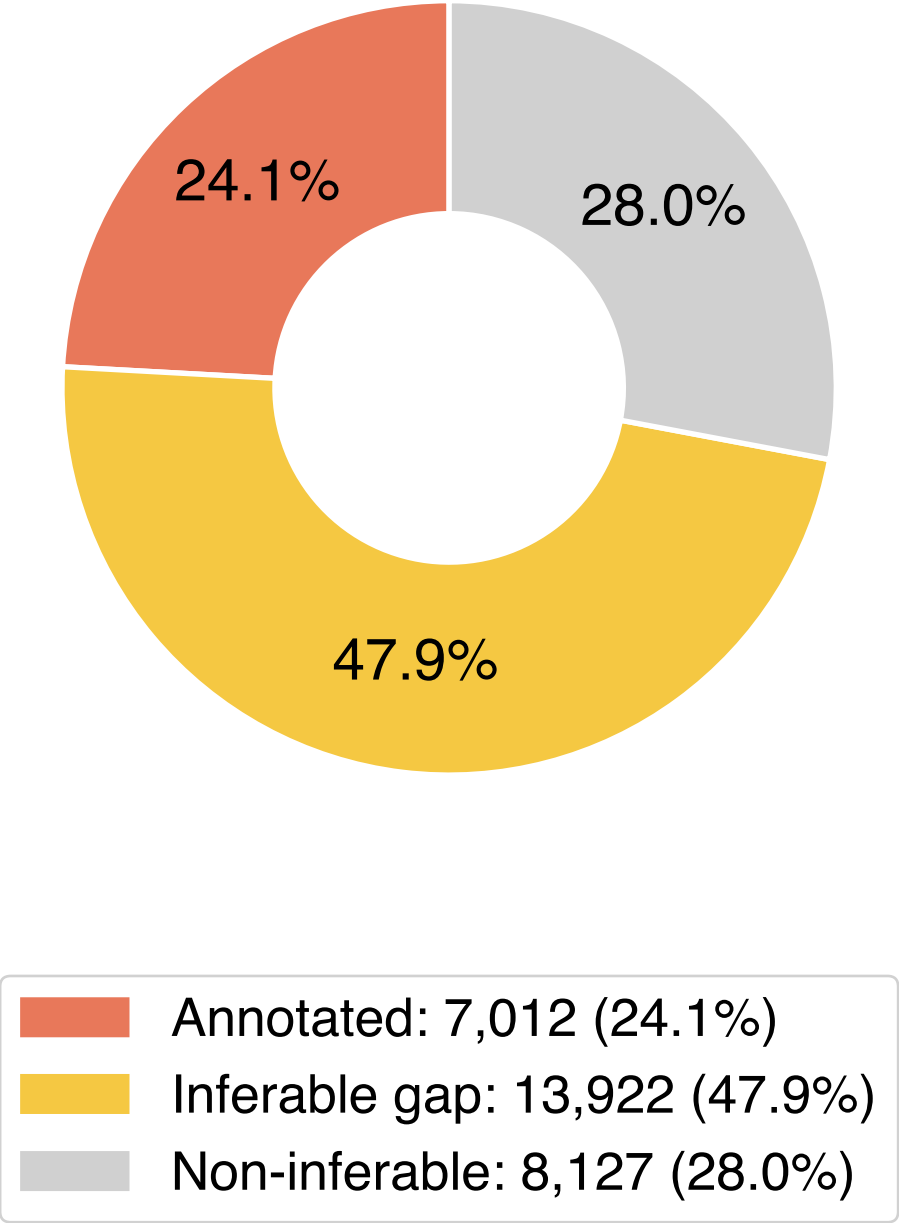


A. Global metabolic annotation
(29,061 reaction × species pairs
across 1,093 pathways)



B. Top 15 pathways by inferable gap size
(numbers = absolute inferable reaction–species count)

